

How Upcoming Games And PPU-Enabled Systems Will Compute And Render A Scene

Let's now look at the same scene being played out in a system that has a CPU, PPU and a GPU (that's a lot of PUs). As usual, data from the game is passed on to the CPU, which sends all graphics computation to the GPU and does all AI computation by itself.

However, it now has the help of the PPU (Physics Processing Unit) to perform complex physics calculations for it

Since the PPU is specifically designed to do physics calculations, it can do a much larger number of calculations in a lot less time than the CPU. In current designs, the PPU will receive data for physics processing from the CPU, but if a game is designed to directly interface with the PPU driver or software, it may bypass the CPU entirely, and this will lead to an even further reduction in the milliseconds of lag that we currently see

The CPU breaks up the data into its respective parts and sends each processing unit its tasks: it computes AI and then passes physics data to the PPU and graphics computation to the GPU for rendering

The PPU then calculates the physics in the scene and sends it to the GPU

The GPU renders the graphics, and then receives AI and physics data from the CPU and PPU respectively—and then renders the whole frame

Because of the PPU's enhanced physics computations, the blast looks more realistic. The same will apply to all in-game fluid dynamics and collisions

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